

Autodesk® Scaleform®

AMP

Scaleform AMP(Analyzer for Memory and Performance)

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: 3.03
: 2013 4 10

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Autodesk® Scaleform® 4.4

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Autodesk Scaleform

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1 소개

AMP™ Scaleform® , Scaleform CPU , , , AMP ActionScript

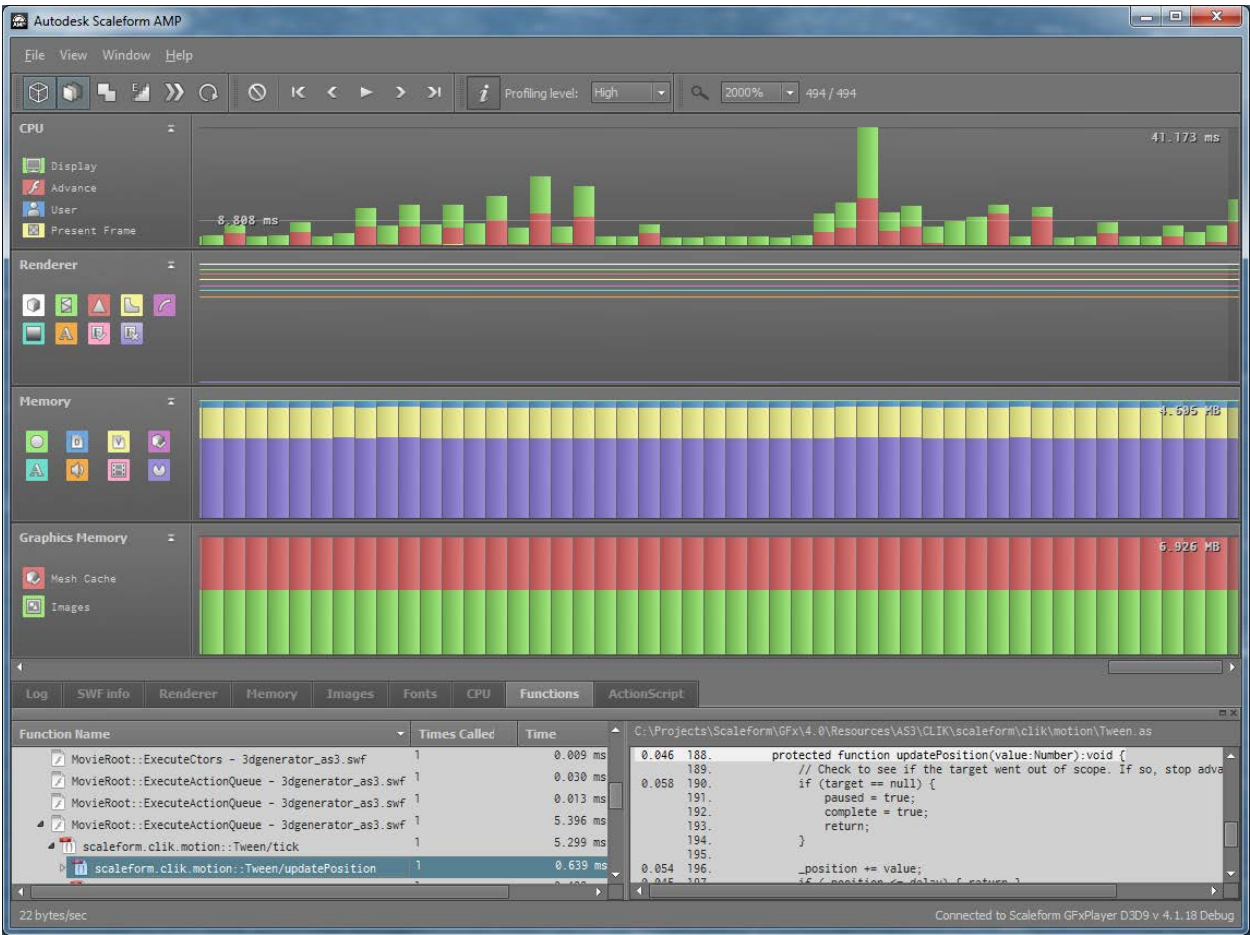


그림 1: Scaleform AMP

Scaleform AMP , AMP AMP

AMP Scaleform 3.2

1.1 AMP

Scaleform (AMP)

Scaleform . ActionScript C++

Scaleform . ActionScript 2

. (ActionScript 3

.) , flash ,

ActionScript .

APM AMP

Shipping AMP ,

AMP .

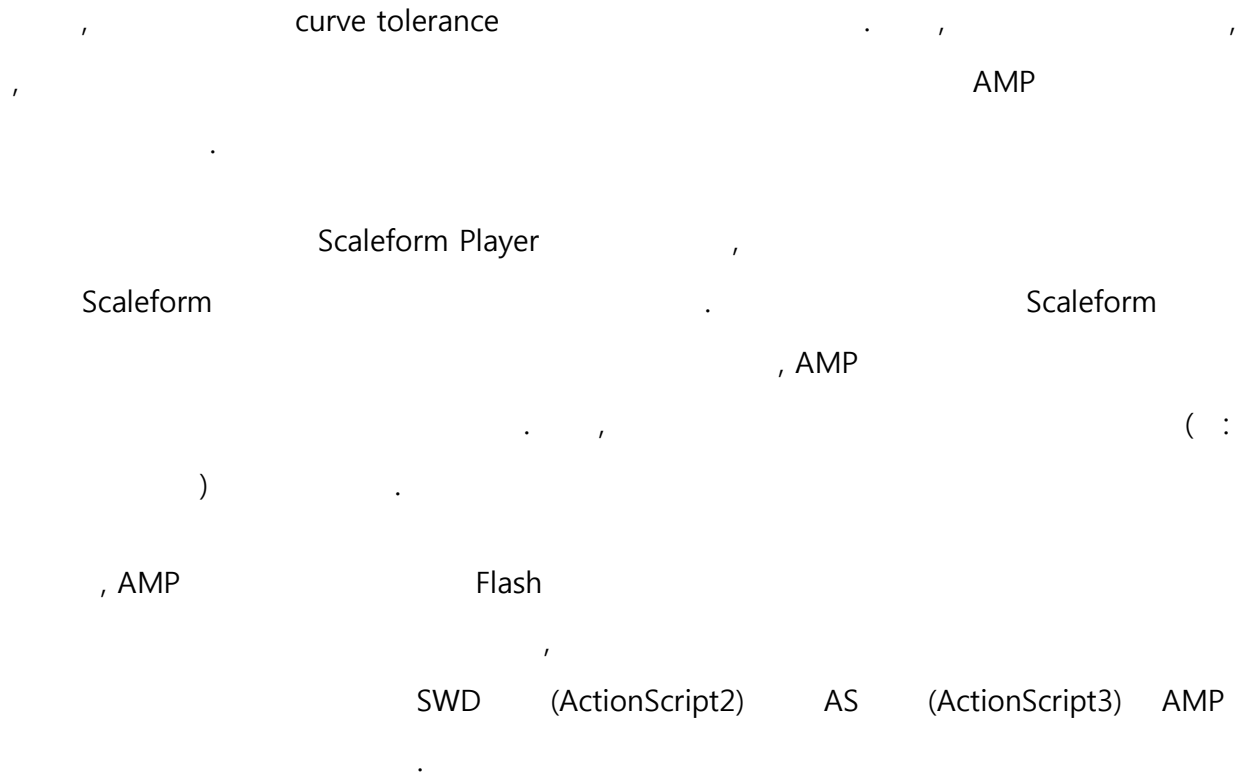
Movie . SWF

view . , AMP

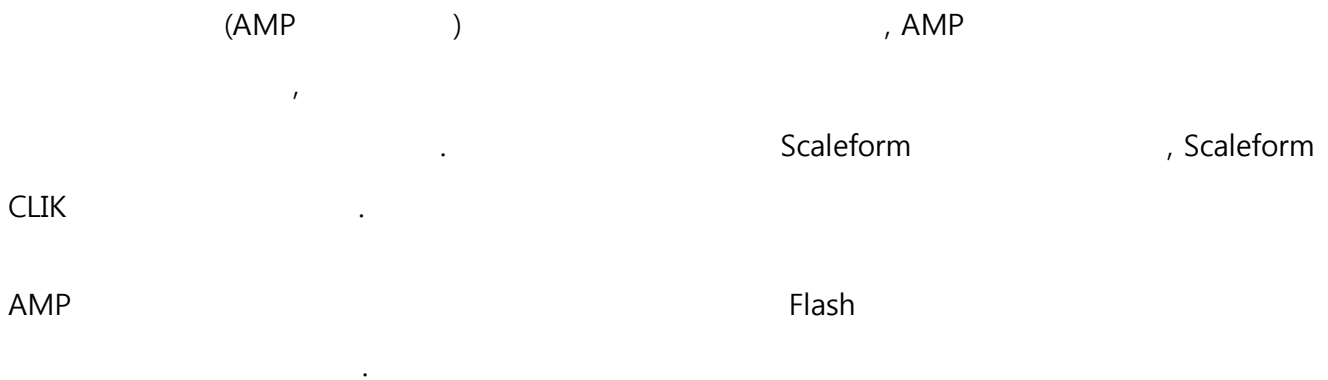
Movies ActionScript .

, AMP

flash movie , , ,



1.2 AMP



- **CPU graph:** ActionScript(Advance), Scaleform (Display), Direct Access API (User), (PresentFrame) CPU

(legend)

(toggle)

- **Rendering graph:** [Best Practices Guide](#)

(primitive),

)

Scaleform

- **Memory graph:** Scaleform

$$\left(\begin{array}{cccc} & & & \\ & & & \\ & & & \\ & & & \end{array} \right)$$

- **Graphics memory graph:** Scaleform

AMP

GPU

- **Log tab:** Log

Scaleform

- **Performance Adviser:** Flash

. Performance Adviser

- o *Text Fields optimized for readability.*

11

11

11

- o *Ratio of draw primitives to number of meshes.*

(1.0)

- o *Number of draw primitives.*

- o *Number of meshes.* ,
- o *Number of triangles.*
- o *Mesh cache thrashing.*
- o *Number of strokes.* " "
- o *Number of gradient fills.* " "
- o *Number of masks.* " "
- o *Number of filters.* " "
- o *Number of DirectAccess API calls.* Direct Access API ,
- o *Number of rasterized glyphs.* ,
- o *Number of font textures.* ,
- o *Number of font texture updates.* .
- o *Font cache thrashing.*
- o *Unused font texture percentage.* ,

- o *Used font cache area:*
- o *Total font cache area:*
- o *Total Memory:* Scaleform
- o *MovieData Memory:*
- o *Font Memory:*
- o *MovieView Memory:* Gfx::Movie
. MovieView
- o *ActionScript Memory:* ActionScript
- o *Mesh Cache Memory:*
- o *Unused Mesh Cache Memory:*
- o *Image Memory:*
- o *Video Memory:* GPU Scaleform
- o *Sound Memory:*
- o *Number of images exceeding a certain size:*

AMP

AMP

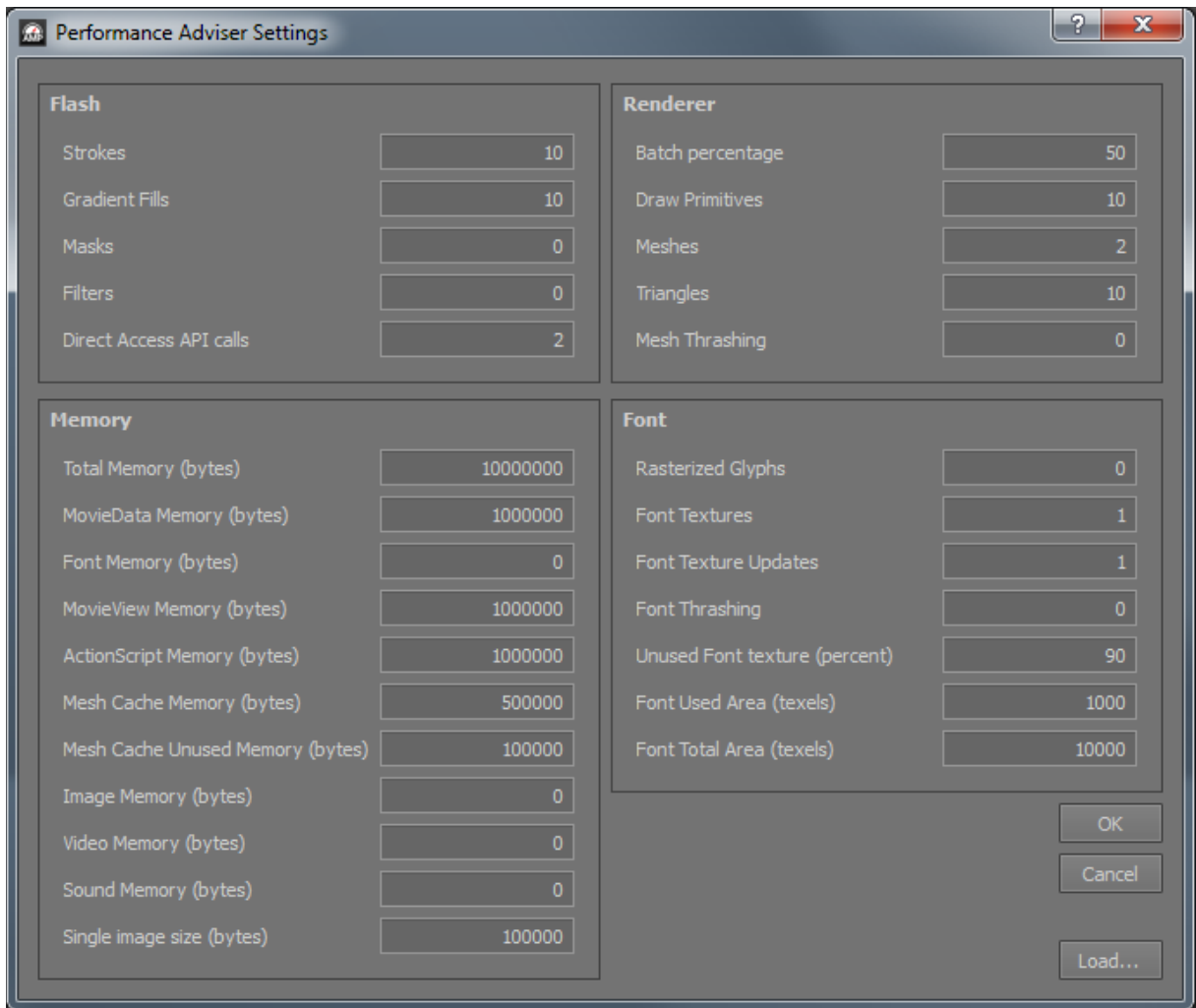
"adviser.xml"

XML

0

Performance Advisor

Settings



OK

AMP

XML

Adviser

"-adviser

<AdviserFilePath>"

AMP

Adviser

- **SWF info tab:**

Frame rate

- **Renderer tab:**

(

),

(

- **Memory tab:** Scaleform, Scaleform
- **Images tab:** Scaleform, AMP, (DXT).
- **Fonts tab:** SWF, Images
- **CPU tab:** Advance, Display, User
- **Functions tab:** . ActionScript C++, ActionScript VM
- **ActionScript tab:** Functions, C++
- **Source pane:** Functions, ActionScript, . AS3, AS, . SWF, AS3, .. AS2, (), " ", . AS2

Scaleform

. 1

SWF
AMP
AS3
flash

- **File Menu:**

- o **Connection:**

- o **Disconnect:**

- o **Debug Info Paths:**

- (AS3)

- Remove

Add

- o **Load Profile Frames:**

- o **Save Profile Frames:**

- "Load Profile Frames"

- o **Dump Memory Report:**

- "AmpMemReport.txt"

- o **Exit:**

- **View menu:**

- o **Graph Tooltips:**

AMP

HW FSAA() None()

- o  **Fast-forward:** flash
SWF
FPS

- o  **Restart:** flash

- **Frame control toolbar:**

- o  **Clear:**
AMP

- o  **First Frame:**

- o  **Previous Frame:**

- o  **Pause:**

- o  **Next Frame:**

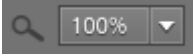
- o  **Last Frame:**

- **Profiling toolbar:**

o  **Detailed Profiling**(): AMP

Memory

AMP Images 10

-  **Zoom toolbar:**

- **Status bar:**

Status Bar

AMP

UI

AMP

Window

Reset UI

CPU CPU Summary, Functions ActionScript
, Function
, ActionScript

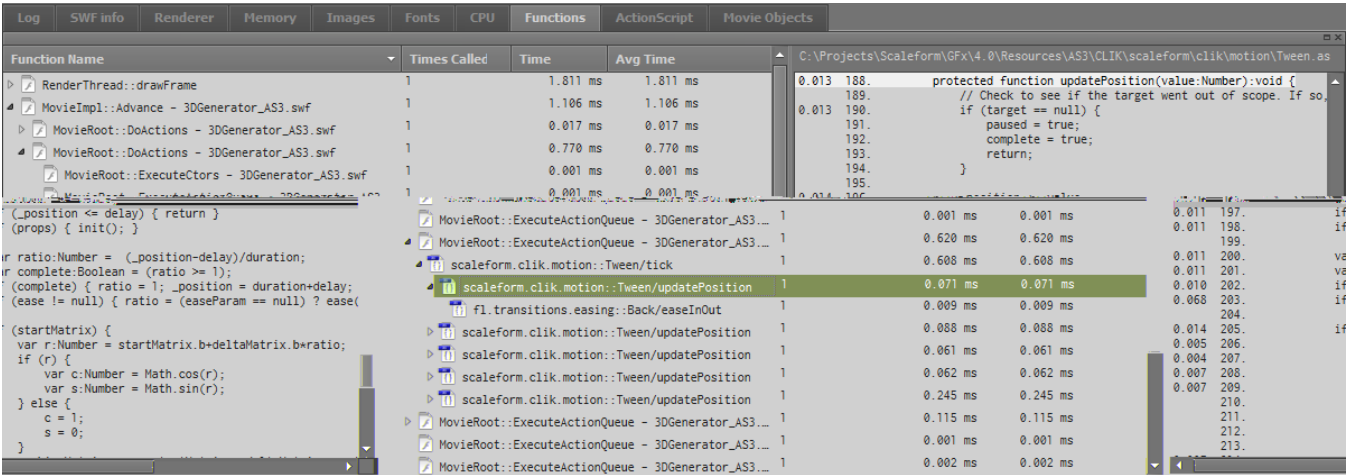


그림 4 : CPU Summary

CPU CPU

2.3

[Best Practices Guide](#)

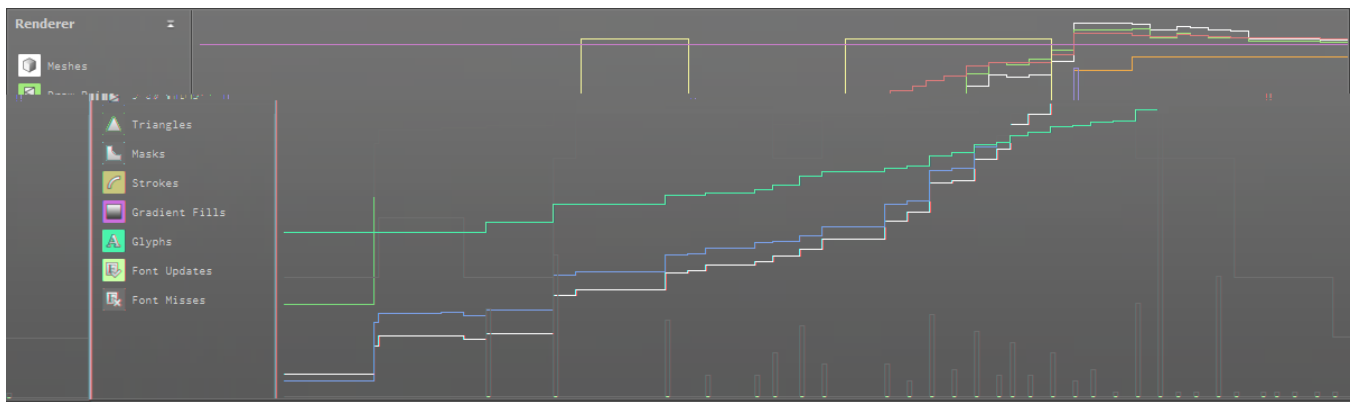


그림 5: 렌더링 그래프

Render Summary

Log	SWF info	Renderer	Memory
Meshe	182		
Draw Primitives:	183		
Triangles:	5124		
Masks:	6		
Filters:	0		
Strokes:	21		
Gradient Fills:	0		
Rasterized Glyphs:	412		
Font Cache Updates:	0		
Font Textures:	1		
Font Cache Thrashing:	0		
Font Cache Fill:	28%		
Font Cache:	1048576 pixels		
Font Cache Used:	298294 pixels		
Font Failures:	0		
Font Cache Misses:	5		
Mesh Thrashing:	0		

그림 6: Rendering Summary

2.4

Scaleform

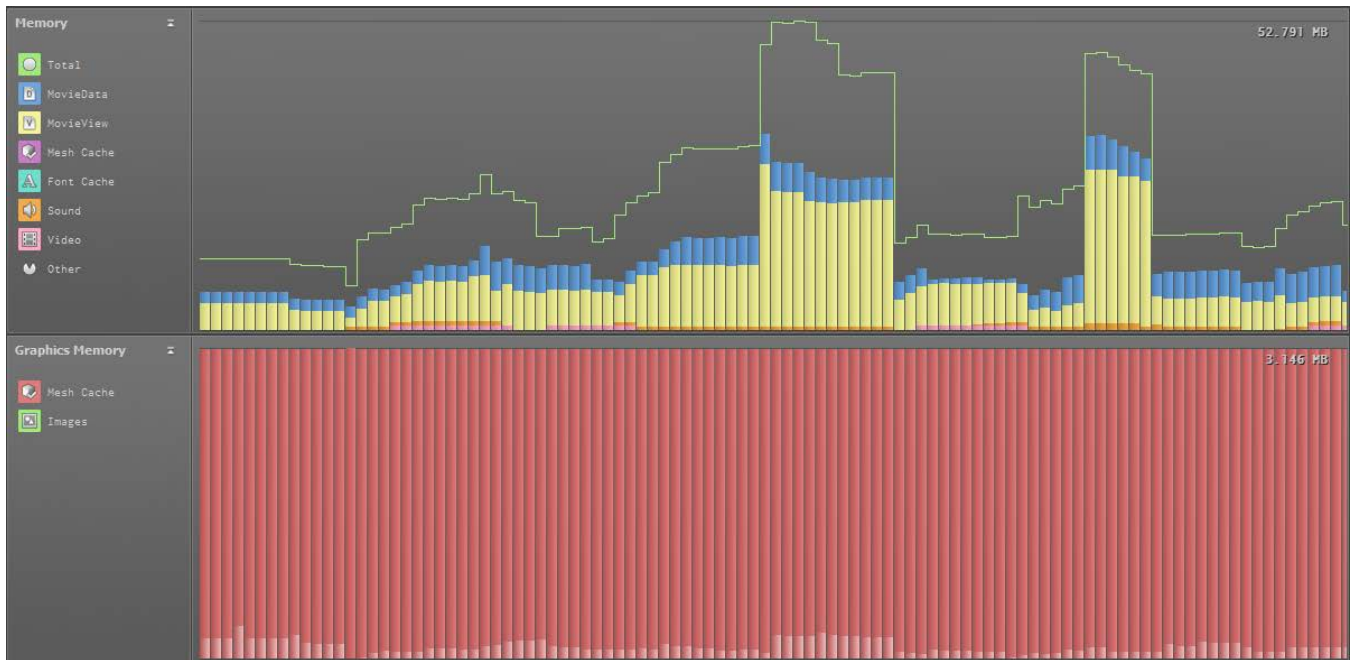


그림 7: Memory 그래프

- **Total:** Scaleform ().
 ,
 .
- **MovieData:** SWF GFX ,
- **Movie view:** , ActionScript
 Gfx::Movie .
- **MeshCache:** ,
 tesselator .
 edge antialiasing .
- **Font Cache:** .

-

-

-

•

“i”

•

5

-

- **Primitive:**
- **Fill:**
- **Mesh:**
- **MeshBatch:** (chunk)
- **Context:**
 - (thread)
- **NodeData:**
- **TreeCache:**
- **TextureManager:**
- **MatrixPool:**
- **MatrixPoolHandles:**
- **Text:** , html (line)
- **Font:**
- **MovieDef:**
 - **CharDefs:**
 - **ShapeData:**
 - **Tags:** ActionScript
 - **Fonts:**

- **Sounds:**
- **ASBinaryData:** ActionScript
- **MD_Other:** MovieDef
- o **MovieView:** ActionScript
 - **MovieClip:**
 - **ActionScript:** ActionScript
 - **ASString:** ActionScript
 - **Text:**
 - **XML:** XML
 - **MV_Other:** Movie View
 - **VM:** ActionScript
 - **AS3 VM:** ActionScript 3
 - **CallFrame:** ActionScript
 - **VTable:** ActionScript
 - **SlotInfo:** ActionScript
 - ClassTraits InstanceTraits
 - SlotInfo ClassTraits
 - InstanceTraits
 - o **SlotInfoHash:** ActionScript (hash)
 - **ClassTraits:** ActionScript
 - **Class:** ActionScript

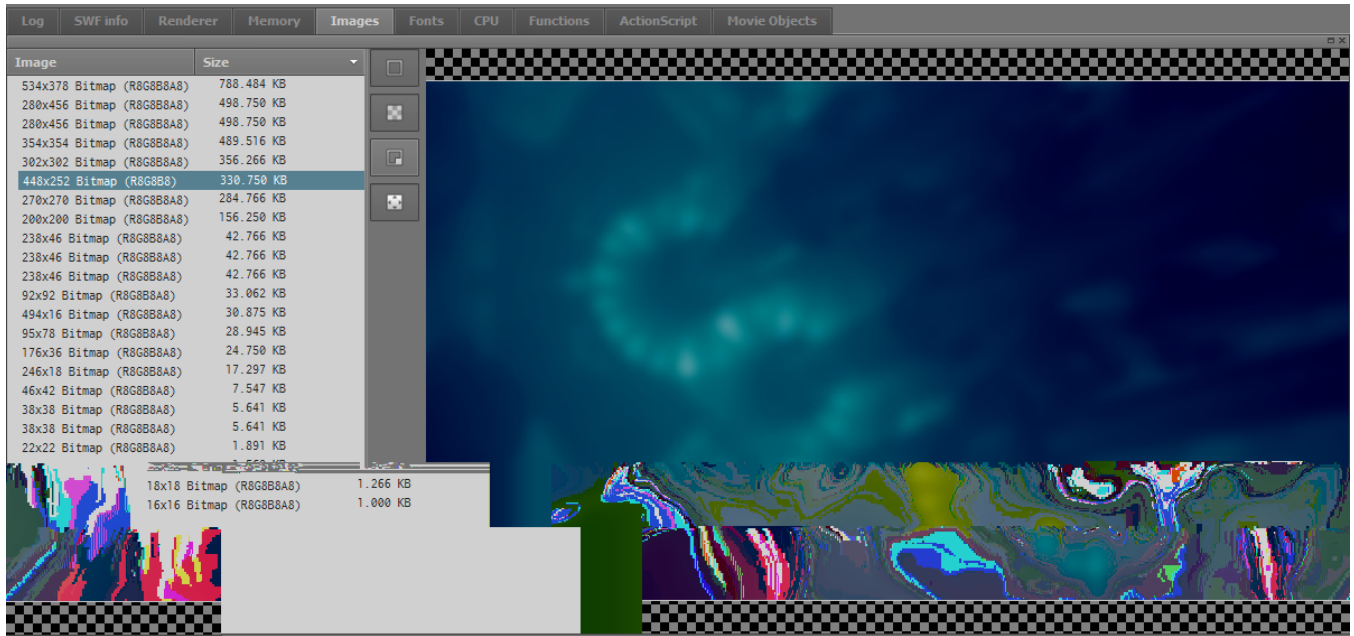
- **InstanceTraits:** ActionScript
- **Instance:** ActionScript
- **AbcFile:**
 - **AbcConstPool:** , , Namespaces Multinames
(constants)
AbcFile
- **VMAbcFile:** AbcFiles ,
- **Tracer:**
- o **IME:** Input Method Editor (IME)

Log	SWF info	Renderer	Memory	Image
Memory				Size
▲	Total Footprint			4849.402 KB
▲	Used Space			4633.715 KB
▷	Video Heaps			0.000 KB
▷	Other Heaps			325.621 KB
▲	Movie View Heaps			1226.094 KB
▲	MovieView "3dgenerator_as3.swf"			1226.094 KB
	String			0.828 KB
▷	Renderer			12.672 KB
▷	MovieView			1176.781 KB
	General			35.812 KB
▷	Movie Data Heaps			261.066 KB
▲	Global Heap			2820.934 KB
	Video			264.176 KB
	String			20.633 KB
	Sound			0.141 KB
▷	Renderer			2310.914 KB
▷	MovieView			8.039 KB
▷	MovieDef			116.676 KB
	Image			0.141 KB
	IME			9.914 KB
	General			90.301 KB
▷	Debug Memory			8.000 KB
▷	Unused Space			38.820 KB
	Debug Data			176.867 KB

그림 8: Memory Summary

Scaleform

Images



크기 9: Images 메모리 탭

Scaleform

..

Scaleform

Fonts



그림 10: Fonts 메모리 탭

SWF

, MovieData

CharDefs

AMP

Heaps

Memory Summary

SWF

2.5

AMP

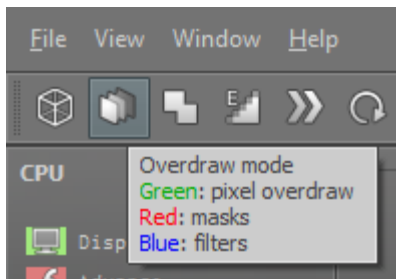


그림 11: 전환 버튼

(16), , (8) . ,

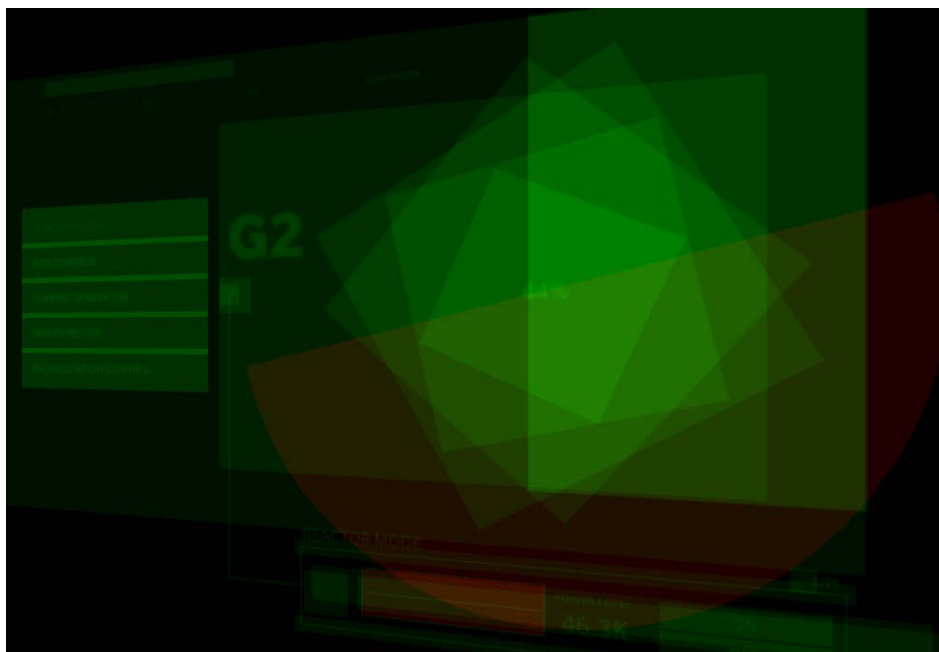


그림 12: Highlighting 을 위한 렌더링 모드

2.6

AMP

Figure 13: Batch profiling mode

3 플랫폼 및 통합 관련 주의사항

- AMP
 - Scaleform::System
 - Scaleform
 - GfX::AMP::Server::Init() GfX::AMP::Server::Uninit()
 - AMP
 - AMP
 - Windows, Xbox360, PS3, Linux, Mac,Wii, Android iPhone/iPad, 3DS, PS Vita, WiiU
 - AMP
 - AMP IP
 - Xbox360 IP
 - Xbox 360 Neighborhood
 - PS3 IP
 - Android
 - AndroidManifest.xml
 - ```
<uses-permission android:name="android.permission.INTERNET" />
```
      - WiFi PC WiiU PC
        - 6003 TCP/IP
      - Wii AMP NDEV
        - USB (COM ) PC IP
        - "wii" Wii AMP
        - : RVL\_SDK/X86/bin hio2.dll
        - AMP
  - Unreal Engine 3
    - AMP
    - Flash
    - SWD/AS3 AS AMP SWF

AS2 SWD

SWD/AS

AMP

.

## 4 빌드 주의사항

- AMP non-shipping Scaleform .  
 Scaleform AMP GfxConfig.h SF\_AMP\_SERVER Gfx  
 . Scaleform  
 AMP . Ptr<Gfx::AMP::ServerState> serverState = \*SF\_NEW  
 Gfx::AMP::ServerState();  
 serverState->StateFlags &= Gfx::AMP::Amp\_Disabled;  
 AmpServer::GetInstance().UpdateState(serverState);
- AMP .  
 (Windows: Ws2\_32.lib,  
 Xbox 360: Xnet.lib, PS3: libnetctl, Linux: ).
- AMP , AMP  
 .  
 .  
 .  
 AMP  
 AMP  
 . AMP::Server::GetInstance().SetInitSocketLib(false).  
 AMP .
- AMP . Scaleform  
 .  
 AMP::Server::GetInstance().SetHeapLimit() .
- AMP 7534 .  
 . AMP::Server::GetInstance().SetListeningPort();



## 6. AMP

```
. AMP::Server::GetInstance().SetConnectionWaitTime.
```



- AMP::Server::GetInstance().SetAppControlCallback  
.
- AMP::Server::GetInstance().SetAppControlCaps AMP  
.  
AMP::MessageAppControl , true  
:  
  
AMP::MessageAppControl caps;  
caps.SetCurveToleranceDown(true);  
caps.SetCurveToleranceUp(true);  
caps.SetNextFont(true);  
caps.SetRestartMovie(true);  
caps.SetToggleAaMode(true);  
caps.SetToggleAmpRecording(true);  
caps.SetToggleFastForward(true);  
caps.SetToggleInstructionProfile(true);  
caps.SetToggleOverdraw(true);  
caps.SetToggleBatch(true);  
caps.SetToggleStrokeType(true);  
caps.SetTogglePause(true);  
caps.SetToggleWireframe(true);  
AMP::Server::GetInstance().SetAppControlCaps(&caps);

### 5.3

AMP

AMP::Server::GetInstance().SetConnectedApp

AMP

- ServerStateType  
AMP::Server::GetInstance().SetState.  
o Amp\_RenderOverdraw

- o Amp\_App\_Wireframe
- o Amp\_App\_Paused
- o Amp\_App\_FastForward
- AMP::Server::GetInstance().SetAaMode
- AMP::Server::GetInstance().SetStrokeType
- AMP::Server::GetInstance().SetCurveTolerance
- AMP::Server::GetInstance().SetCurveTolerance
- AMP::ServerState
- AMP::Server::GetInstance().UpdateState

## 5.4

AMP C++ ActionScript CPU  
 . ActionScript

Amp.addMarker(1)

: Amp.addMarker() .as

Resources/AS2/CLIK

Amp

C++

GFx::MovieImpl:: AdvanceStats->AddMarker(1)

AMP

## 6 AMP 관련 FAQ

AMP

### 1. 게임 측에서 AMP 를 사용하는 데 필요한 최저 사양은 무엇입니까?

GFX\_AMP\_SERVER

AMP

GfX

AMP

. Platform::RenderThread::drawFrame

AmpServer::GetInstance().AdvanceFrame

AMP

### 2. AmpClient 를 사용하여 게임을 연결하면 게임 프레임률이 현저하게 떨어집니다. 조사해보니 AmpServer::AdvanceFrame()을 실행하면 시간이 오래 걸리는데 이게 정상입니까?

AMP

AMP

"I"( I)

. AMP

### 3. Windows/PS3/Xbox 에서 게임을 실행합니다. 검색창에서 AMP 클라이언트를 검색해 AMP 에 연결할 수는 있습니다. 그런데 통계 정보가 전혀 없습니다.

AMP

Platform::RenderThread::drawFrame

AmpServer::GetInstance().AdvanceFrame

AMP

### 4. (AMP::Server::WaitForAmpConnection 을) 업데이트할 때 왜 AMP 가 일 초 이상 멈춥니까?

AMP

AMP

Scaleform

AMP

'i'

. AMP

'profile level'

1MB

GFx::AMP::Server::SetHeapLimit

**5. 프로파일러를 사용할 때 예상되는 메모리 오버헤드는 어느 정도입니까?**

AMP

1MB

AmpServer::SetHeapLimit

AMP

GFx

## 7 추가 정보

AMP

- [AMP Use Cases](#) .
- [Scaleform Reference](#) – – /
- [AMP Forum](#) –